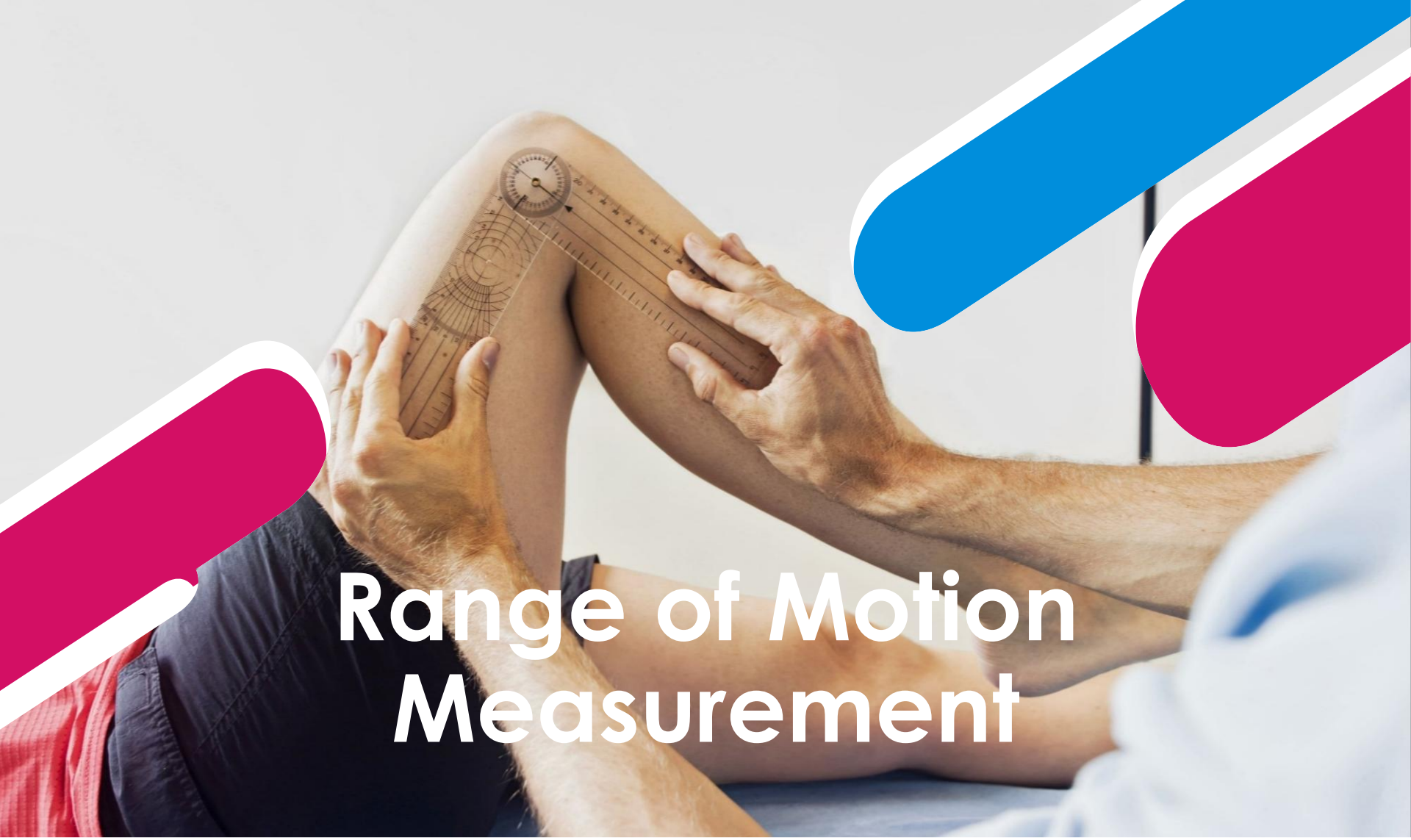


Range of Motion Measurement

Walaa Abu-Taleb

Learning Outcomes

By the end of this session, you should be able to:

- Describe planes of movement and the axes of rotation of the human musculoskeletal system.
- Understand the difference between active and passive ROM and indications
- List the types of end feel in joints.
- Accurately measure the range of movement in selected joints using goniometry and other tools.
- List some of the objective measurement of Range of motion.

Steps in the MSK

Assessment

1- Interview and history taking (subjective assessment)

2- Objective Assessment:

1. Observation and Palpation
2. ROM Ax: Active, passive and Accessory movement
3. Muscle testing
4. Gait Ax
5. Balance Testing
6. Neuro Ax : Neuro-integrity and neuro-dynamics (neural provocation tests)
7. Functional Ax

DATE:
NAME:
AGE:
OCCUPATION:

HISTORY

Mechanism of Injury:

Aggravating/Easing Factors or Movements:

24-Hour History:

Improving/Static/Worse

New/Old Injury

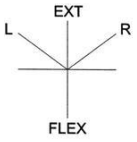
Past History (include social and family history):

Diagnostic Imaging:

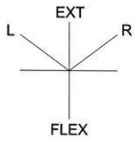
OBSERVATION (POSTURE)

EXAMINATION

ACTIVE MOVEMENTS

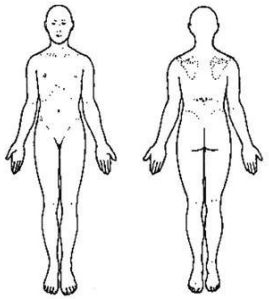


PASSIVE MOVEMENTS



Comments:
End Feel:
Capsular Pattern:

Mark where symptoms are:



VAS: Intensity of Pain

no pain 0 — pain as bad as it could possibly get 10

Pain: constant, periodic, episodic, occasional

RESISTED ISOMETRIC MOVEMENTS

Comments:

FUNCTIONAL TESTING

NEUROLOGICAL TESTS

Sensory Scan:
Reflexes:
Neurological Special Tests:

SPECIAL TESTS

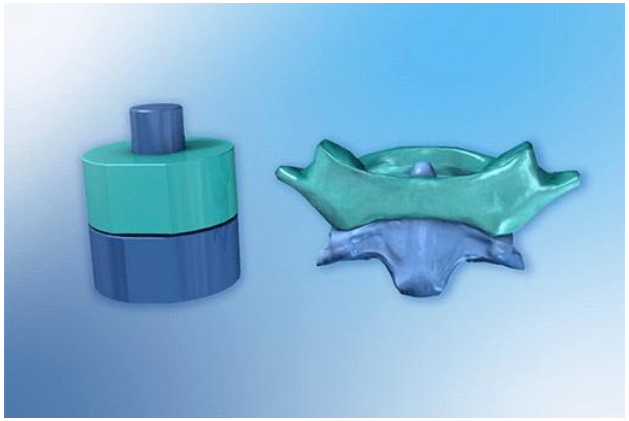
JOINT PLAY MOVEMENTS

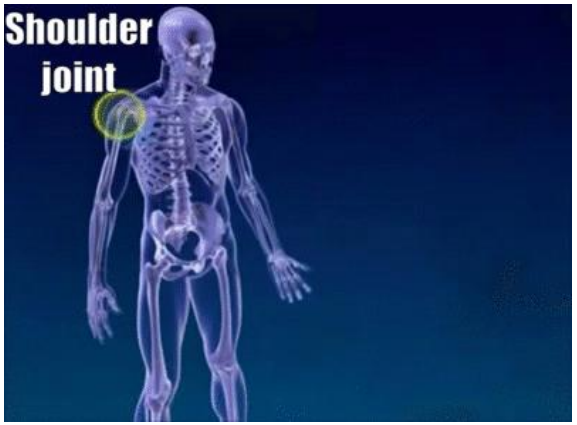
PALPATION

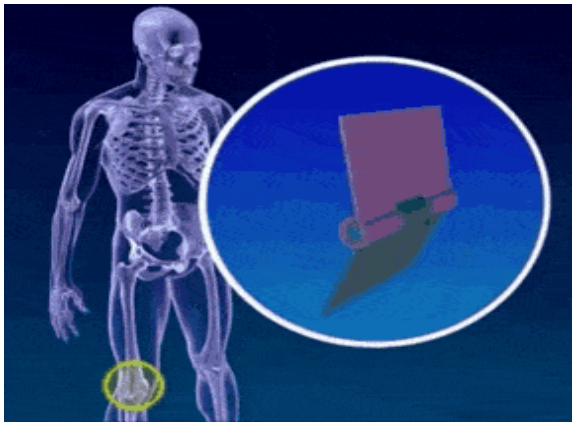
Tenderness, Effusion

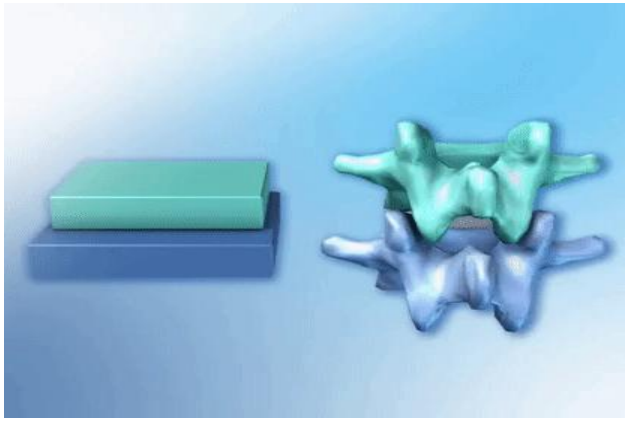
(use reverse side for other comments)

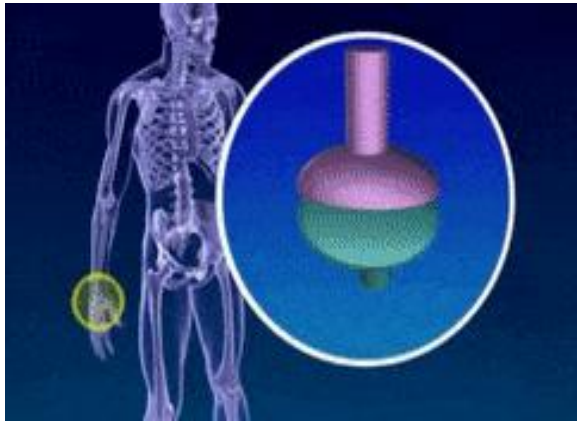
Types of Synovial Joints

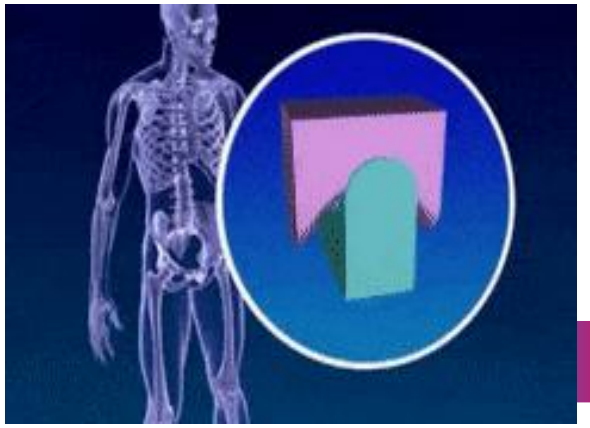






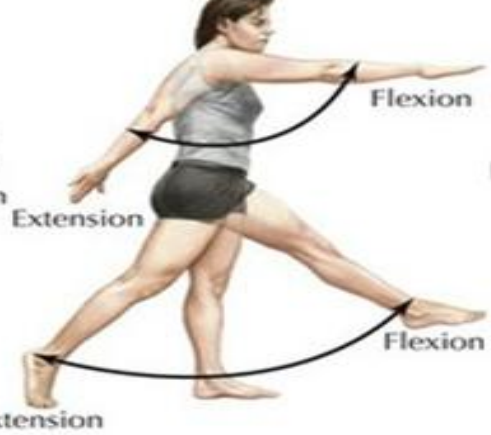


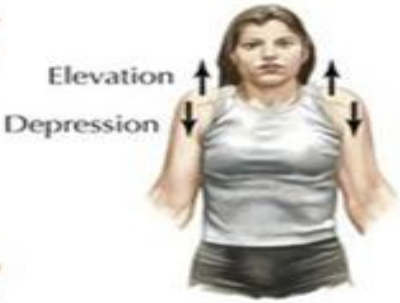


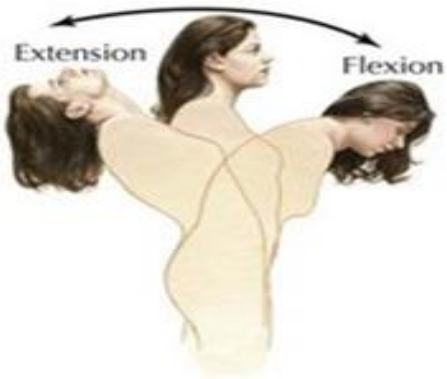


Extremities

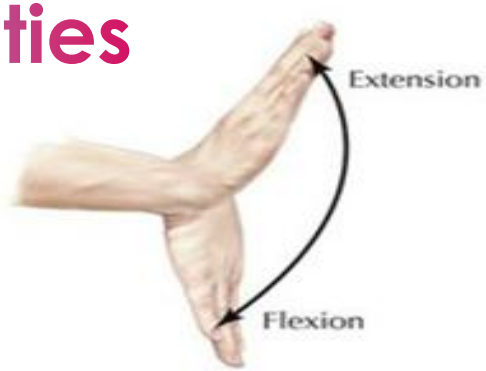






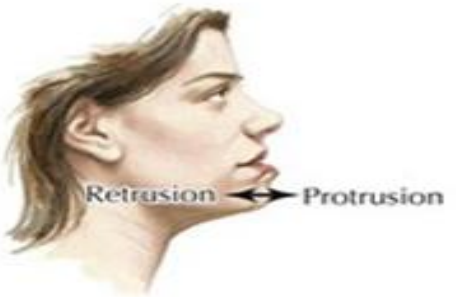


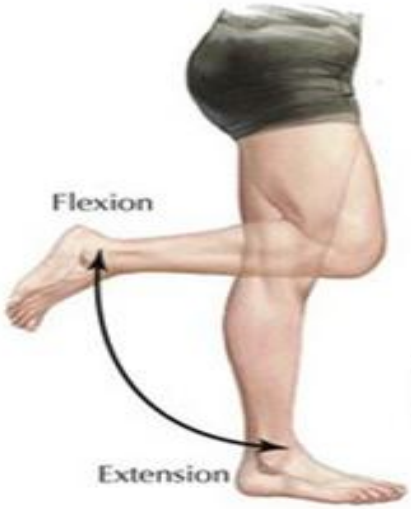






C. Machado M.D.

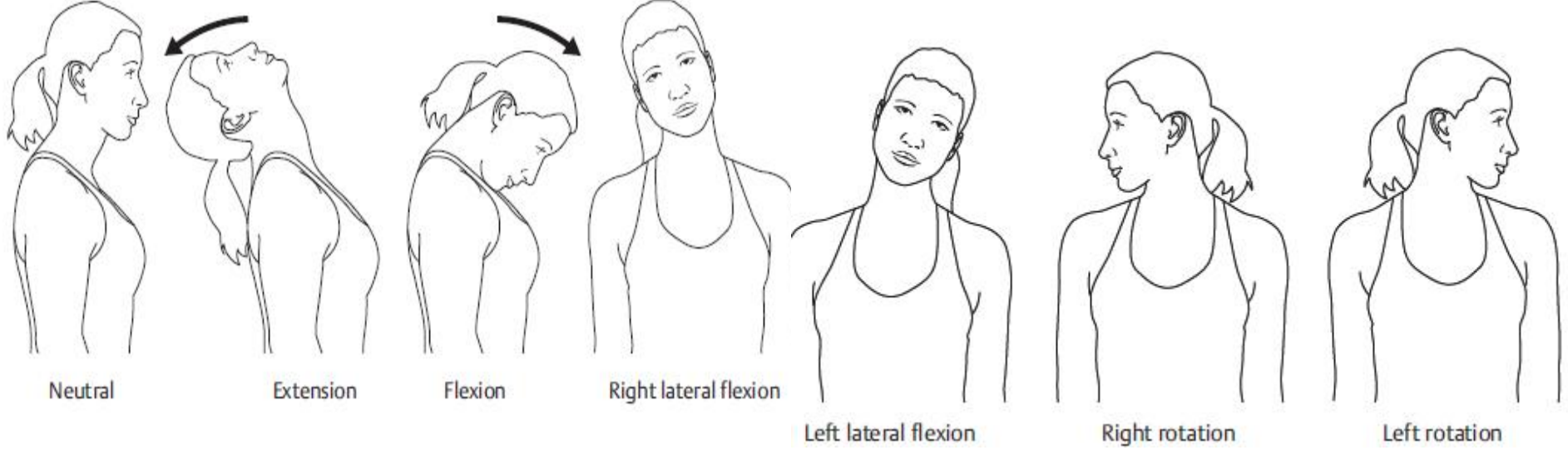




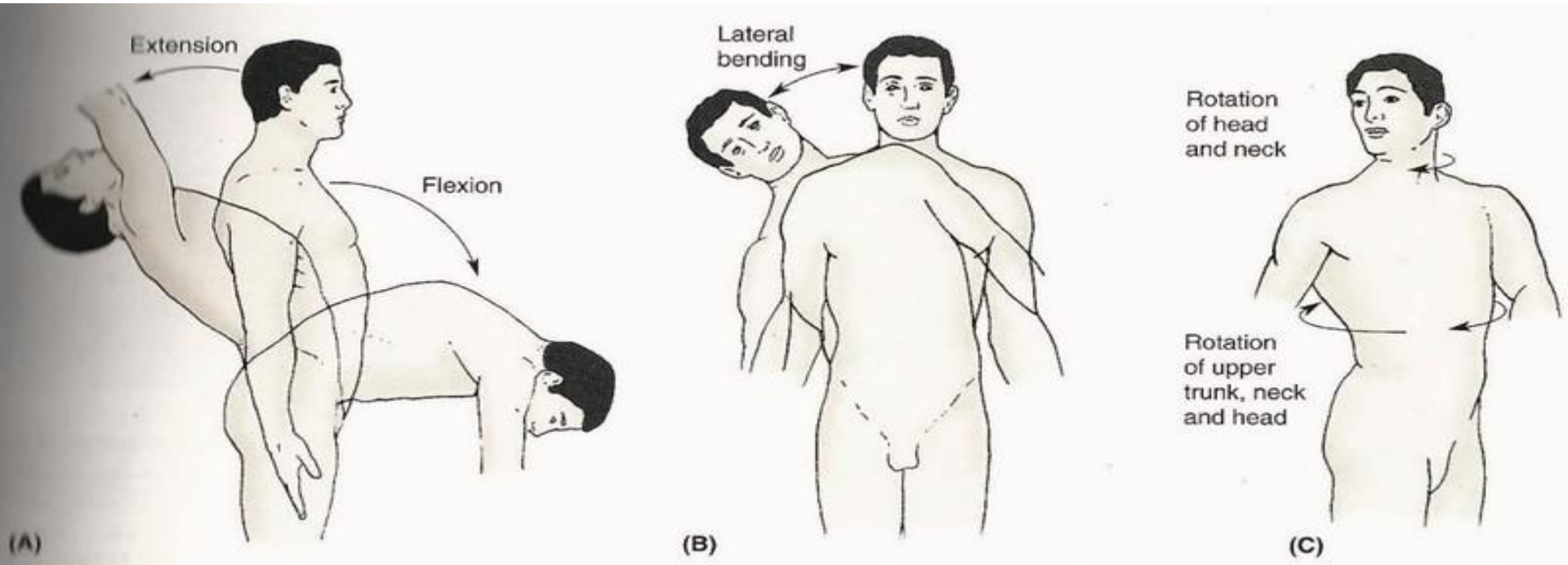






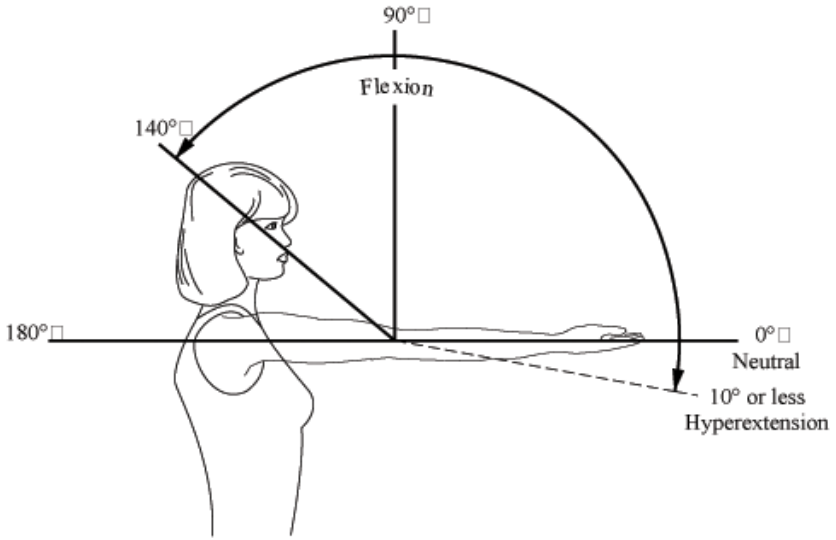

Neutral
Extension
Flexion
Right lateral flexion
Left lateral flexion
Right rotation
Left rotation

Spinal


Extension
Flexion
(A)
Lateral bending
(B)
Rotation of head and neck
Rotation of upper trunk, neck and head
(C)

Range of motion (ROM)

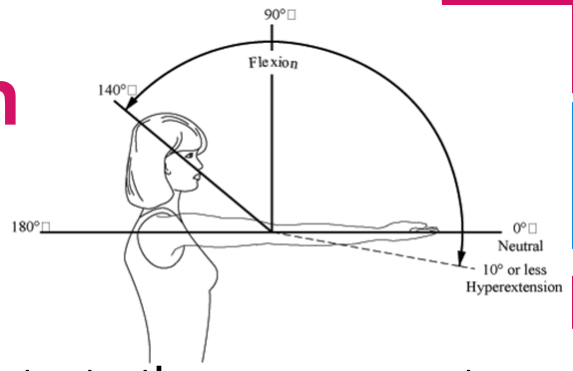
- ROM is the amount of motion that can occur between any two bones in any movable joint.
- It is a technique used for the examination of movement and for initiating movement into a program of therapeutic intervention.



Use it or Lose it

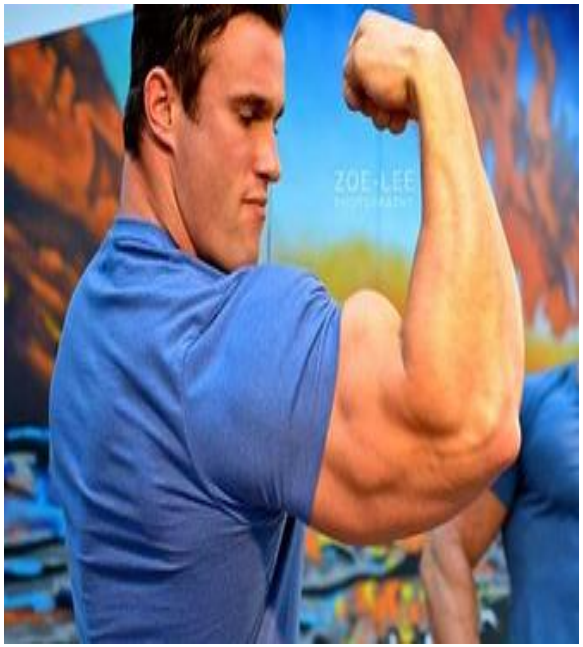
	Joint	Normal Range of Motion (Degrees)
Shoulder	Flexion	0-180
	Extension	0-60
	Abduction	0-180
	Internal rotation	0-70
	External rotation	0-90†
Elbow	Flexion	0-150
Forearm	Pronation	0-80
	Supination	0-80
Wrist	Flexion	0-80
	Extension	0-70
Hip	Flexion	0-120
	Extension	0-30
	Abduction	0-45
	Adduction	0-30
	Internal rotation	0-45
	External rotation	0-45
Knee	Flexion	0-135
Ankle	Dorsiflexion	0-20
	Plantar flexion	0-50

Principles of Range of Motion (AROM) Assessment



- Assess the environment before the patient starts the movement to ensure their safety.
- Ensure the patient understands what you will do and what to expect.
- Explain any expectations of pain and to inform you when it occurs and location.
- Ensure the patient is supported and not at risk of falling or losing balance or at a risk of a limb drop.
- The patient should be comfortable, and respected as it may influence the ROM achieved. Also, the patient needs to be wearing comfortable close to not restrict the movement.
- Understand the factors that influence the measurement of ROM and include in your clinical reasoning.

Confounding factors of ROM Assessment

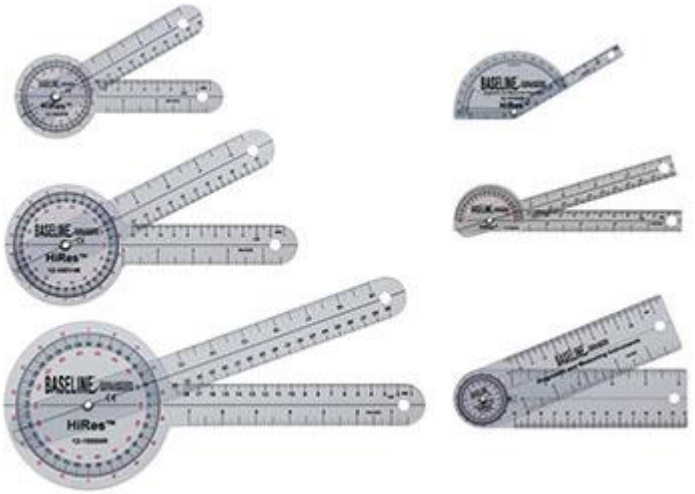


- Age.
- Gender.
- Weight.
- Dominant versus Non-Dominant Side (Muscle Size).
- Compensation.
- Test Position.
- Test Procedures.
- Tool used.

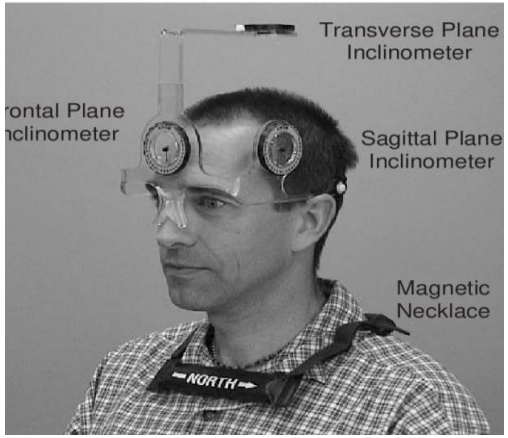
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Tools to measure ROM

- Subjective Measures: Visual Estimation
- Objective measures:
- Goniometers and inclinometers
- Cervical: CROM
- Lumbar BROM
- Mobile Apps: Eyeglass type wearable sensor (JINS MEME ES_R)









Types of ROM

- Passive ROM (PROM)
- Active ROM (AROM)
- Active-Assistive ROM(A-AROM)/ Auto-Assisted ROM

Principles of Testing Active ROM

- The movement is performed actively throughout the ROM.
- The therapist observes and asks questions including symptoms during the movement and the limiting factor at the end of the available range.
- ROM is then measured.
- Appy a gentle pressure at the end of the range to assess the (end feel).
- Records the findings



In ROM Assessment, the therapist observes:



- Rhythm and quality of movement
- Symmetry between the RT and LT sides.
- Any restrictions of movement and location
- Ease of movement
- Willingness of the patient to move
- Movement of associated joints or any substitutions: Example: Shoulder elevation or trunk side bending with flexion or abduction.

If there is pain during movement

The therapist enquires:

- When and where the onset of pain is
- Whether movement increases the intensity and quality of pain
- Reaction of the patient to pain
- Signs of Kinesiophobia behaviour.

2- Principles of Passive ROM Assessment

The limb is carried safely and passively moved by the therapist.

The joint is allowed to move freely through the range.

The tested part must be supported throughout the motion then placed back to the resting position.

The patient's face is also observed and communicate with the patient about their symptoms.

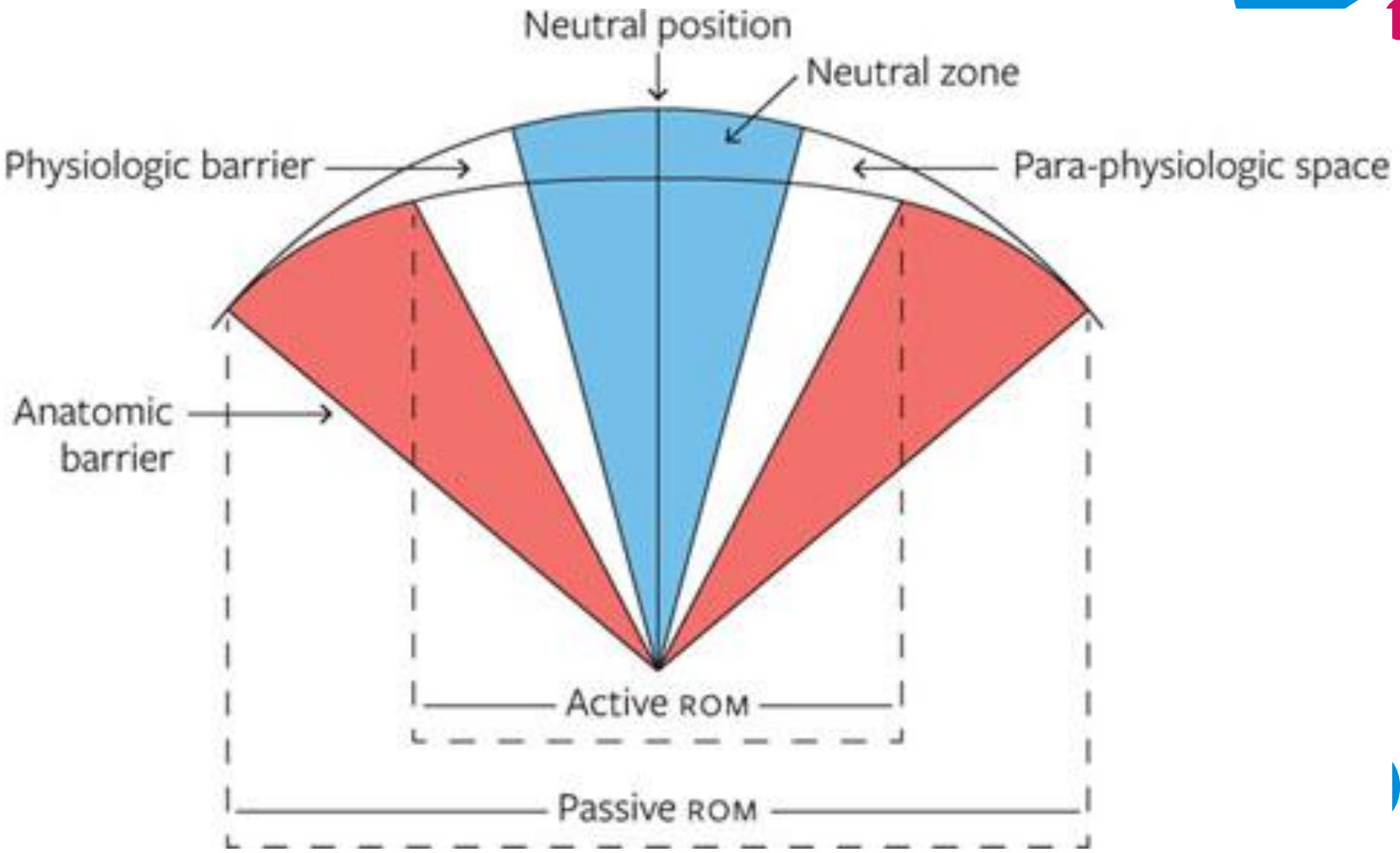
At the end of the movement, note end feeling and limiting factors (Pain/ Stiffness, etc).

The ROM is then measured.

Record the findings.



Active and Passive ROM and Anatomic barrier



Precaution

- Pain
- Osteoporosis or bone fragility as a forced measurements may cause iatrogenic injury
- Hypermobility or Instability
- Some illnesses as: AS or Haemophilia
- After prolonged immobilisation

Contraindications

- If the body part will disrupt a healing process if moved such as unhealed or unstable fracture, rupture of a tendon / ligament, infectious or acute inflammatory process, myositis ossificans or ectopic ossification, suspected or confirmed DVT.

Preparation of Assessment

1. Check for contraindications or precautions.
2. Organise the testing sequence by body position to minimise changes in positioning.
3. Communication and Consent to expose part and touch.
4. Positioning: Ensure the patient is comfortable and well-supported. The joint can move through its full ROM without obstruction.
5. Avoid active and passive insufficiency
6. Stabilisation: proximal joint is stabilised to minimise any substitute movements
7. Assess End Feel
8. Alignment of the Measurement Tool:
    - Fulcrum or Axis - Positioned over a point near the joint's axis of rotation.
    - Stationary Arm - Usually aligned with the midline of the stationary segment of the joint.
    - Moving Arm - Usually aligned with the midline of the moving segment of the joint.

Procedure

- Stabilization
- Fixation
- Position of patient and therapist
- Movement
- Patient’s order



Digital Halo goniometer

End-feel	Description	Example
Soft	•Soft Tissue ApproximationOccurs when two soft tissue masses meet one another, limiting further movement. •The quality of resistance is soft, and this gradually increases as the soft tissue is compressed between body parts.	Knee Flexion: Contact between the soft tissue of the posterior leg and posterior thigh
Firm	•Muscular End FeelOccurs when muscular tension limits the ROM. •Quality of resistance is firm, although not as firm as with a capsular end-feel, and somewhat springy. •Feels like stretching a tyre inner tube	Hip Flexion with Knee Straight (SLR):Passive elastic tension of the hamstring muscles
	•Ligamentous End FeelOccurs when tension in the ligaments surrounding the joint limits the ROM. •Quality of resistance felt is firm, although not as firm as with capsular end-feel, and somewhat springy. •Feels like stretching a leather belt.	Forearm Supination: Tension in the palmar radioulnar ligament of the inferior radioulnar joint, interosseous membrane and oblique cord
	•Capsular End FeelOccurs when the joint capsule and surrounding non-contractile tissues limit the ROM. •Quality of resistance felt is firm but not hard. There is a slight "give" to the movement. •Feels like stretching a leather belt, with more resistance than a ligament.	Extension of Metacarpophalangeal Joints: Tension in the anterior capsule
Hard	•Bone on Bone: Occurs when the approximation of two bones stops the ROM. •Quality of the resistance felt is very hard and abrupt, with further motion impossible.	Elbow Extension; Contact between the olecranon process of the ulna and the olecranon fossa of the humerus

Demonstration and practice



1. Elbow flexion and extension
2. Hip External and Internal Rotation



Now you practice…



What are findings of assessment?

What are you looking for?

- ROM – quantitative measure (degrees)
- Endfeel
- Quality of movement (through range)
- Symptom response through range
- Limiting factor (and symptom response)

Hypermobility

Beighton score

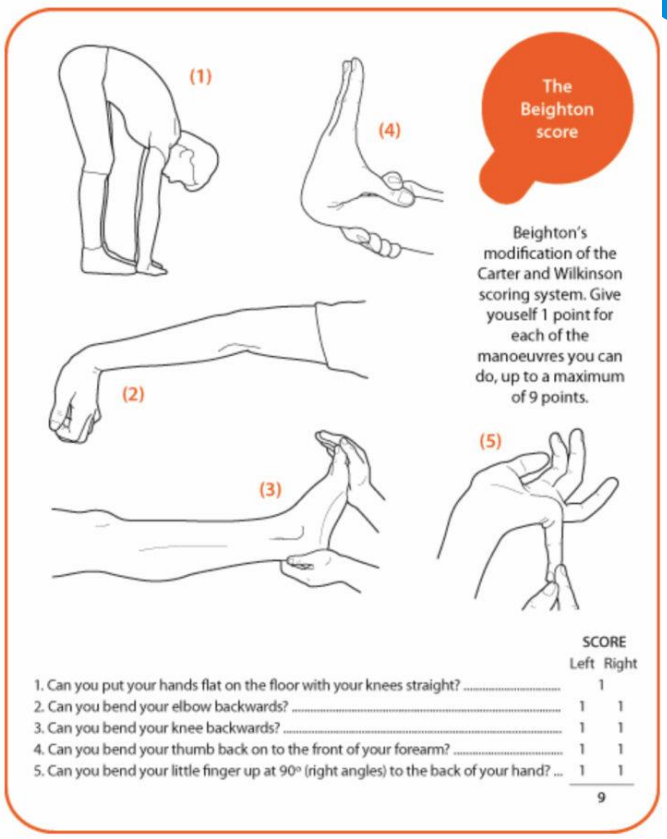












Source: Arthritis Research UK

Beighton Score

Movement	LEFT	RIGHT
1. Passive dorsiflexion and hyperextension of the fifth MCP joint beyond 90°	1	1
2. Passive apposition of the thumb to the flexor aspect of the forearm	1	1
3. Passive hyperextension of the elbow beyond 10°	1	1
4. Passive hyperextension of the knee beyond 10°	1	1
5. Active forward flexion of the trunk with the knees fully extended so that the palms of the hands rest flat on the floor	1	
TOTAL	/9	